

Binned K -Propagation for Coordinate-Spiked ReLU Networks

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1 Setting and motivation

Model. Let f be a depth- L ReLU MLP with square hidden layers of width n ,

$$X^{(0)} \sim \mathcal{N}(0, I_n), \quad Z^{(\ell)} = M^{(\ell)} X^{(\ell-1)} + b^{(\ell)}, \quad X^{(\ell)} = \rho(Z^{(\ell)}),$$

followed by a linear readout. Each hidden matrix carries a coordinate spike on the axis $e = e_1$:

$$M = W + e_1 e_1^\top, \quad W_{ij} \sim \mathcal{N}(0, 1/n).$$

The target is the output mean $\mathbb{E}[f(X)]$.

Spike/bulk split. Every layer vector is decomposed along the spike axis,

$$X = A e + B, \quad A = e^\top X \in \mathbb{R}, \quad B = \Pi X \in e^\perp, \quad \Pi = I - e e^\top, \quad (1)$$

where A is the scalar *spike coordinate* and B the d -dimensional *bulk*, $d = n - 1$. Relative to $\mathbb{R}^n = \mathbb{R}e \oplus e^\perp$ a hidden matrix block-decomposes as

$$M = \begin{pmatrix} \gamma & r^\top \\ u & V \end{pmatrix}, \quad \gamma = M_{00}, \quad r = M_{0,1:}, \quad u = M_{1:,0}, \quad V = M_{1:,1:},$$

so one linear layer acts as

$$\boxed{A^+ = \gamma A + r^\top B, \quad B^+ = A u + V B.} \quad (2)$$

2 State

The joint law of a layer vector X is approximated by the mixture

$$\hat{\mathcal{L}}(X) = \sum_{\alpha=1}^m p_\alpha \delta_{v_\alpha}(dA) \otimes \nu_\alpha(dB), \quad \nu_\alpha \approx \mathcal{N}(\mu_\alpha, \Sigma_\alpha), \quad (3)$$

stored as four arrays:

symbol	shape	meaning
p_α	$(m,)$	$P(A \in \text{bin } \alpha)$; nonnegative, sums to 1
v_α	$(m,)$	representative spike value $\approx \mathbb{E}[A \mid \text{bin } \alpha]$
μ_α	(m, d)	conditional bulk mean $\mathbb{E}[B \mid \text{bin } \alpha]$
Σ_α	(m, d, d)	conditional bulk covariance $\text{Cov}[B \mid \text{bin } \alpha]$

Coordinate 0 never appears inside μ or Σ ; the spike lives only in (p, v) . Within a bin, A is collapsed to the single point v_α .

3 Initialization

TLDR: We bin the initial state, notations ϕ and Φ denote the Gaussian pdf and cdf respectively.

With $X^{(0)} \sim \mathcal{N}(0, I_n)$ we have $A^{(0)} \sim \mathcal{N}(0, 1)$ independent of $B^{(0)} \sim \mathcal{N}(0, I_d)$. Given pre-activation edges $z_0 = -\infty < z_1 < \dots < z_m = +\infty$ defining bins $[z_{\alpha-1}, z_\alpha)$,

$$p_\alpha = \Phi(z_\alpha) - \Phi(z_{\alpha-1}), \quad v_\alpha = \mathbb{E}[A^{(0)} \mid \text{bin } \alpha] = \frac{\varphi(z_{\alpha-1}) - \varphi(z_\alpha)}{\Phi(z_\alpha) - \Phi(z_{\alpha-1})}, \quad \mu_\alpha = 0, \quad \Sigma_\alpha = I_d, \quad (4)$$

where φ, Φ are the standard-normal pdf and cdf. The truncated-normal mean v_α is already the W_2 -optimal representative for the cell (Section 7).

4 Linear step

Fix an old bin α : inside it, $A = v_\alpha$ is a point and $B \sim \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$. Push both through (2).

4.1 The joint Gaussian (Y, C)

TLDR: Calculating the mean and covariance of one of the binned perpendicular Gaussians X pass a linear layer.

The new scalar spike pre-activation and new bulk vector are one affine map of the Gaussian B ,

$$\begin{pmatrix} Y_\alpha \\ C_\alpha \end{pmatrix} = \begin{pmatrix} \gamma \\ u \end{pmatrix} v_\alpha + \begin{pmatrix} r^\top \\ V \end{pmatrix} B,$$

hence jointly Gaussian, with mean and covariance read off directly:

$$\begin{pmatrix} Y_\alpha \\ C_\alpha \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} m_Y \\ m_C \end{pmatrix} = \begin{pmatrix} \gamma \\ u \end{pmatrix} v_\alpha + \begin{pmatrix} r^\top \\ V \end{pmatrix} \mu_\alpha, \quad \begin{pmatrix} s_Y^2 & g^\top \\ g & \Sigma_C \end{pmatrix} = \begin{pmatrix} r^\top \\ V \end{pmatrix} \Sigma_\alpha \begin{pmatrix} r^\top \\ V \end{pmatrix}^\top\right). \quad (5)$$

4.2 Transition kernel

For each new bin $I_\beta = [\ell_\beta, h_\beta)$, with standardized limits $a = (\ell_\beta - m_Y)/s_Y$ and $b = (h_\beta - m_Y)/s_Y$,

$$Q_{\beta\alpha} = P(Y_\alpha \in I_\beta) = \Phi(b) - \Phi(a).$$

New bin masses follow by marginalization, and Bayes gives the posterior of the old bin given the new:

$$p_\beta^+ = \sum_\alpha p_\alpha Q_{\beta\alpha} \quad (p^+ = Qp), \quad \eta_{\alpha|\beta} = \frac{p_\alpha Q_{\beta\alpha}}{p_\beta^+}. \quad (6)$$

This is exactly one step of a hidden-Markov filter with emission “which bin did Y land in.”

4.3 Conditioning the bulk on $Y \in I_\beta$

TLDR: Conditional mean and covariance of the transformed gaussian MX_α from bin α to bin β . Formula in 7

The truncated-normal moments of Y on I_β (writing $Z = Q_{\beta\alpha}$) are

$$\tau_1 = \mathbb{E}[Y - m_Y \mid I_\beta] = s_Y \frac{\varphi(a) - \varphi(b)}{Z}, \quad \tau_2 = \mathbb{E}[(Y - m_Y)^2 \mid I_\beta] = s_Y^2 \left[1 + \frac{a\varphi(a) - b\varphi(b)}{Z} \right],$$

so $\text{Var}(Y \mid I_\beta) = \tau_2 - \tau_1^2$.

To condition C on where Y landed, split C into the part carried by Y and a remainder independent of Y . With $\kappa := g/s_Y^2$, define

$$C_\perp := C_\alpha - m_C - \kappa(Y_\alpha - m_Y).$$

The coefficient κ is chosen exactly so that the remainder is uncorrelated with Y : $\text{Cov}(C_\perp, Y) = g - \kappa s_Y^2 = 0$; and since (Y, C) is jointly Gaussian, uncorrelated means *independent*. $C_\perp$ is thus a zero-mean Gaussian, independent of Y , with

$$\text{Cov}(C_\perp) = \Sigma_C - 2\kappa g^\top + \kappa \kappa^\top s_Y^2 = \Sigma_C - \frac{gg^\top}{s_Y^2}.$$

Now condition the decomposition on the event $Y \in I_\beta$. Because $C_\perp$ is independent of Y , the conditioning only touches the $\kappa(Y - m_Y)$ term, and mean/covariance of the two independent pieces add:

$$\mathbb{E}[C \mid Y \in I_\beta] = m_C + \kappa \mathbb{E}[Y - m_Y \mid I_\beta] + 0 = m_C + \kappa \tau_1,$$

$$\text{Cov}[C \mid Y \in I_\beta] = \kappa \kappa^\top \text{Var}(Y \mid I_\beta) + \text{Cov}(C_\perp),$$

i.e.

$$\boxed{\mu_{\alpha \rightarrow \beta} = m_C + \frac{g}{s_Y^2} \tau_1, \quad \Sigma_{\alpha \rightarrow \beta} = \Sigma_C - \frac{gg^\top}{s_Y^2} + \frac{gg^\top}{s_Y^4} (\tau_2 - \tau_1^2).} \quad (7)$$

There is a small degenerate case to deal with. When $s_Y^2 \approx 0$: Y_α is the constant m_Y , so it lands wholly in the bin containing m_Y , with $\mu_{\alpha \rightarrow \beta} = m_C$ and $\Sigma_{\alpha \rightarrow \beta} = \Sigma_C$ (no division by s_Y^2).

4.4 Mixture over old bins

Several old bins feed each new bin β ; they combine with the posterior weights (6) by the laws of total mean and total covariance:

$$v_\beta^+ = \sum_\alpha \eta_{\alpha|\beta} \mathbb{E}[Y_\alpha \mid I_\beta] = \sum_\alpha \eta_{\alpha|\beta} (m_{Y,\alpha} + \tau_{1,\alpha\beta}), \quad (8)$$

$$\mu_\beta^+ = \sum_\alpha \eta_{\alpha|\beta} \mu_{\alpha \rightarrow \beta}, \quad \Sigma_\beta^+ = \sum_\alpha \eta_{\alpha|\beta} \left[\Sigma_{\alpha \rightarrow \beta} + (\mu_{\alpha \rightarrow \beta} - \mu_\beta^+) (\mu_{\alpha \rightarrow \beta} - \mu_\beta^+)^\top \right]. \quad (9)$$

5 ReLU step

ReLU is coordinatewise, so it respects the split (1): $\rho(X) = \rho(A) e + \rho(B)$.

5.1 Bulk, inside each bin

TLDR: Calculating the mean and covariance of the ReLU-ed Gaussian within each bin. Note this is $O(n^2)$.

With $B \sim \mathcal{N}(\mu_\alpha, \Sigma_\alpha)$, $\sigma_i^2 = \Sigma_{\alpha,ii}$, $z_i = \mu_i/\sigma_i$, the exact rectified-Gaussian marginal moments are

$$\mathbb{E}[\rho(B_i)] = \mu_i \Phi(z_i) + \sigma_i \varphi(z_i), \quad \mathbb{E}[\rho(B_i)^2] = (\mu_i^2 + \sigma_i^2) \Phi(z_i) + \mu_i \sigma_i \varphi(z_i),$$

giving $\tilde{\mu}_{\alpha,i} = \mathbb{E}[\rho(B_i)]$ and $\tilde{\Sigma}_{\alpha,ii} = \mathbb{E}[\rho(B_i)^2] - \tilde{\mu}_{\alpha,i}^2$. Off-diagonal entries use the exact bivariate-Gaussian ReLU covariance $\text{Cov}(\rho(B_i), \rho(B_j))$ via Owen's T . With correlation $\theta = \Sigma_{\alpha,ij}/(\sigma_i \sigma_j)$, $s = \sqrt{1 - \theta^2}$, and

$$P = \Phi_2(z_i, z_j; \theta), \quad q = \varphi_2(z_i, z_j; \theta), \quad a_i = \varphi(z_i) \Phi\left(\frac{z_j - \theta z_i}{s}\right), \quad a_j = \varphi(z_j) \Phi\left(\frac{z_i - \theta z_j}{s}\right),$$

the closed form is

$$\mathbb{E}[\rho(B_i)\rho(B_j)] = \mu_i \mu_j P + \mu_j \sigma_i (a_i + \theta a_j) + \mu_i \sigma_j (a_j + \theta a_i) + \sigma_i \sigma_j [\theta P + (1 - \theta^2) q - \theta z_i a_i - \theta z_j a_j], \quad (10)$$

Importantly, this depends only on the mean and covariance restricted to the 2 coordinates, thus $O(1)$ to calculate.

Implementation details: The backend choice "kprop" delegates to the ordinary harmonic-kprop ReLU.

5.2 Spike and bin merge

ReLU output is nonnegative, so the representatives map $\tilde{v}_\alpha = \max(v_\alpha, 0)$ onto a *nonnegative* post-activation grid with a dedicated zero bin. Old bins landing in the same post bin β , $S_\beta = \{\alpha : \text{bin}(\tilde{v}_\alpha) = \beta\}$, merge by mixture moments with $\eta_{\alpha|\beta} = p_\alpha/p_\beta^+$:

$$p_\beta^+ = \sum_{\alpha \in S_\beta} p_\alpha, \quad v_\beta^+ = \sum_{\alpha \in S_\beta} \eta_{\alpha|\beta} \tilde{v}_\alpha, \quad \mu_\beta^+ = \sum_{\alpha \in S_\beta} \eta_{\alpha|\beta} \tilde{\mu}_\alpha, \quad \Sigma_\beta^+ = \sum_{\alpha \in S_\beta} \eta_{\alpha|\beta} [\tilde{\Sigma}_\alpha + (\tilde{\mu}_\alpha - \mu_\beta^+)(\tilde{\mu}_\alpha - \mu_\beta^+)^T]. \quad (11)$$

This bin is a bit special when trying to generalize to the assumption of mean and covariance being linear in e_1 direction, because it combines a pile of previously normal bins. Thus we always track the 0 bin separately.

6 Readout and recovered moments

A full network alternates LINEAR_STEP (with pre-activation edges, which must include negative bins since $A^+ = \gamma A + r^\top B$ can be negative even when the incoming $A \geq 0$) and RELU_STEP (with nonnegative post edges). The readout is linear, so after the last hidden ReLU the full mean is reconstructed as

$$\mathbb{E}[X] = \bar{A} e + \bar{B}, \quad \bar{A} = \sum_{\alpha} p_{\alpha} v_{\alpha}, \quad \bar{B} = \sum_{\alpha} p_{\alpha} \mu_{\alpha}, \quad (12)$$

and the output mean is $W_{\text{ro}} \mathbb{E}[X] + b_{\text{ro}}$. The full covariance, when needed, follows by the law of total covariance over the finite spike law.

Algorithm 1: BINNED-KPROP ($K = 2$, m spike bins)

1. $(p, v, \mu, \Sigma) \leftarrow \text{GAUSSIANINITIALSTATE}(d, \text{edges})$: truncated-normal p_α, v_α ; $\mu = 0, \Sigma = I_d$.
 2. For each hidden layer with matrix $M = \begin{pmatrix} \gamma & r^\top \\ u & V \end{pmatrix}$ and bias:
 - (a) **Linear step**: for each old bin α compute $(m_Y, s_Y^2, m_C, \Sigma_C, g)$ by (5);
 - (b) transition kernel $Q_{\beta\alpha}$, masses $p^+ = Qp$, posteriors $\eta_{\alpha|\beta}$ (eq. (6))
 - (c) condition the bulk on $Y \in I_\beta$: $(\mu_{\alpha \rightarrow \beta}, \Sigma_{\alpha \rightarrow \beta})$ (eq. (7))
 - (d) find new bins and merge over old bins: (v^+, μ^+, Σ^+) (eqs. (8)–(9))
 - (e) **ReLU step**: per bin, exact rectified-Gaussian bulk update $(\tilde{\mu}_\alpha, \tilde{\Sigma}_\alpha)$; $\tilde{v}_\alpha = \max(v_\alpha, 0)$;
 - (f) combine negative bins at 0 bin.
 3. Return $W_{\text{ro}}(\bar{A}e + \bar{B}) + b_{\text{ro}}$ with $\bar{A} = \sum_\alpha p_\alpha v_\alpha$, $\bar{B} = \sum_\alpha p_\alpha \mu_\alpha$.
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7 Bin placement by Wasserstein-2 quantization

7.1 Minimizing mean difference

Binning replaces the continuous law of A by the m -atom law $\hat{A} = \sum_\alpha p_\alpha \delta_{v_\alpha}$. Within a cell every value of A is reported as v_α , so the squared error between the true and binned mean is

$$\sum_\alpha \int_{\text{cell } \alpha} (a - v_\alpha)^2 f(a) da = \sum_\alpha p_\alpha \text{Var}(A \mid \text{cell } \alpha) = W_2^2(\mathcal{L}(A), \hat{\mathcal{L}}(A)), \quad (13)$$

as there are no closed form objective, for choosing the bins, to minimize the algorithm's prediction error, I imagine choosing the bins to minimize the distance between 2 distributions to be a good enough surrogate. Empirically error tends to stabilize after around $o(n)$ bins regardless of how I chose the bins.

7.2 Lloyd–Max optimality conditions

Minimizing (13) over both partition and representatives yields the two stationarity conditions of optimal scalar quantization: (i) representatives are cell centroids, $v_\alpha = \mathbb{E}[A \mid \text{cell } \alpha]$; (ii) edges are midpoints of adjacent representatives, $z_\alpha = \frac{1}{2}(v_\alpha + v_{\alpha+1})$. Alternating (i) and (ii) is Lloyd's algorithm; for log-concave densities (Gaussian) it converges to the global optimum.

7.3 The exact per-layer law of A is a Gaussian mixture

Under the $K = 2$ closure the expected continuous law of the next spike coordinate is available in closed form, and that is what gets quantized:

- **Pre-activation grid.** A^+ is exactly a Gaussian mixture, one component per current bin:

$$\mathcal{L}(A^+) = \sum_\alpha p_\alpha \mathcal{N}(m_{Y,\alpha}, s_{Y,\alpha}^2), \quad m_{Y,\alpha} = \gamma v_\alpha + r^\top \mu_\alpha, \quad s_{Y,\alpha}^2 = r^\top \Sigma_\alpha r.$$

The cell centroid of a mixture is closed form: with $\alpha_k = (a - m_k)/s_k$, $\beta_k = (b - m_k)/s_k$,

$$\mathbb{E}[A^+ | [a, b]] = \frac{\sum_k w_k [m_k(\Phi(\beta_k) - \Phi(\alpha_k)) + s_k(\varphi(\alpha_k) - \varphi(\beta_k))]}{\sum_k w_k (\Phi(\beta_k) - \Phi(\alpha_k))}.$$

This coincides with the linear step's representative v_β^+ in (8), so the propagated per-bin value is the *exact mixture centroid*, never a single-Gaussian approximation.

- **Post-activation grid.** Same bins as pre-activation, but the negative bins all merge at 0.

Two grid builders exist: `lloyd_max_edges` (single moment-matched Gaussian, cheap) and `lloyd_max_edges_mixture` (the true mixture, closed-form centroids each iteration). On a 3-component test the mixture version cut W_2^2 to the true law by 14–28% over the moment-matched one.

A Closed forms used

Standard normal $\varphi(x) = e^{-x^2/2}/\sqrt{2\pi}$, Φ its cdf. Truncated normal on $[a, b]$ with $Z = \Phi(b) - \Phi(a)$ (and $x\varphi(x) \rightarrow 0$ as $x \rightarrow \pm\infty$):

$$\mathbb{E}[A | a \leq A < b] = \frac{\varphi(a) - \varphi(b)}{Z}, \quad \text{Var} = 1 + \frac{a\varphi(a) - b\varphi(b)}{Z} - \left(\frac{\varphi(a) - \varphi(b)}{Z}\right)^2.$$

Rectified Gaussian, $Y \sim \mathcal{N}(\mu, \sigma^2)$, $z = \mu/\sigma$:

$$\mathbb{E}[\rho(Y)] = \mu\Phi(z) + \sigma\varphi(z), \quad \mathbb{E}[\rho(Y)^2] = (\mu^2 + \sigma^2)\Phi(z) + \mu\sigma\varphi(z), \quad P(Y \leq 0) = \Phi(-z).$$

B Implementation map

The $K = 2$ core (`Mecha_preds/binning_kprop/core.py`) is numpy/scipy and torch-free: `BinnedK2State`, `gaussian_initial_state`, `linear_step_k2`, `relu_step_k2`, `run_binned_kprop_k2`, `unconditional_mean[_cov]`; Gaussian-ReLU integrals come from the shared kernel `Mecha_preds/_utils`. Grid builders live in `binning.py` (`make_gaussian_edges`, `make_relu_post_edges`, `lloyd_max_edges[_mixture][_split]`). `kprop_hook.py` bridges to ordinary harmonic kprop for the "kprop" bulk-ReLU backend and the $K > 2$ per-bin tower. `adapter.py` provides the drop-in `run_binned_kprop(model, config={"num_bins": m})`; `selftest.py`, `scaling.py`, `benchmark.py` verify against Monte Carlo, check the n^{-2} law, and tune m respectively.

[implementation in /Mecha_preds/binning_kprop](#)